Matrix Determinant

Let $A = (a_{ij})_{n\times n}$ be a square matrix of order n , then the number $|A|$ called determinant of the matrix A.

(i) Determinant of 2×2 matrix

Let A= $\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ then $|A| = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$

(ii) Determinant of 3×3 matrix

Let B = $\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$

Then $|B| = a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$

$$|B| = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) - a_{13}(a_{21}a_{32} - a_{31}a_{22})$$

Exercise 3.1: Calculate the determinants of the following matrices

(i) $A = \begin{pmatrix} 1 & 3 & 4 \\ 2 & 6 & 8 \\ 1 & 9 & 5 \end{pmatrix}$ (ii) $B = \begin{pmatrix} 2 & -3 & 4 \\ 5 & 6 & 7 \\ 8 & 9 & 1 \end{pmatrix}$

Minor and Cofactor of a Matrix

Let A = $(a_{ij})_{n\times n}$ is a square matrix. Then M_{ij} denote a sub matrix of A with order (n-1) × (n-1) obtained by deleting its ith row and j^{th} column. The determinant $|M_{ij}|$ is called the minor of the element a_{ij} of A.

The cofactor of a_{ij} denoted by A_{ij} and is equal to $(-1)^{i+j}|M_{ij}|$.

Exercise 3.2: Let $A = \begin{pmatrix} 5 & 4 & 2 \\ 2 & 3 & 1 \\ 3 & -2 & -1 \end{pmatrix}$

(i) Compute determinant of A.

(ii) Find the cofactor matrix.

Adjoint Matrix

The transpose of the matrix of cofactors of the element a_{ij} of A denoted by $adj\ A$ is called adjoin of matrix A.

Inverse of a Matrix

If A and B are two matrices such that $AB = BA = I$, then each is said to be inverse of the other. The inverse of A is denoted by A^{-1}.

Example

- $A = \begin{bmatrix} 5 & 4 & 2 \\ 2 & 3 & 1 \\ 3 & -2 & -1 \end{bmatrix}$

i. Compute determinant of A.

ii. Find the cofactor matrix of A.

iii. Find the adjoint matrix of A.

iv. Find the inverse of the matrix A

(i) Compute determinant of A.

- $A = \begin{bmatrix} 5 & 4 & 2 \\ 2 & 3 & 1 \\ 3 & -2 & -1 \end{bmatrix}$
- $|A| = 5\begin{vmatrix} 3 & 1 \\ -2 & -1 \end{vmatrix} - 4\begin{vmatrix} 2 & 1 \\ 3 & -1 \end{vmatrix} + 2\begin{vmatrix} 2 & 3 \\ 3 & -2 \end{vmatrix}$
- $= 5(-3-(-2)) - 4(-2-3) + 2(-4-9)$

  $= 5\times(-1) - 4\times(-5) + 2\times(-13)$

$= -5 + 20 - 26 = -11$

(ii)Find the cofactor matrix of A.

$$C = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}$$

- $C_{ij} = (-1)^{i+j}\left|M_{ij}\right|$
- $C_{11} = (-1)^{1+1}|M_{11}| = (-1)^{2}\begin{vmatrix} 3 & 1 \\ -2 & -1 \end{vmatrix} = 1\times(-3+2) = -1$
- $C_{12} = (-1)^{1+2}|M_{12}| = (-1)^{3}\begin{vmatrix} 2 & 1 \\ 3 & -1 \end{vmatrix} = -1\times(-2-3) = 5$
- $C_{13} = (-1)^{1+3}|M_{13}| = (-1)^{4}\begin{vmatrix} 2 & 3 \\ 3 & -2 \end{vmatrix} = 1\times(-4-9) = -13$
- $C_{21} = (-1)^{2+1}|M_{21}| = (-1)^{3}\begin{vmatrix} 4 & 2 \\ -2 & -1 \end{vmatrix} = -1\times(-4+4) = 0$
- $C_{22} = (-1)^{2+2}|M_{22}| = (-1)^{4}\begin{vmatrix} 5 & 2 \\ 3 & -1 \end{vmatrix} = 1\times(-5-6) = -11$

- $C_{23} = (-1)^{2+3}|M_{23}| = (-1)^5 \begin{vmatrix} 5 & 4 \\ 3 & -2 \end{vmatrix} = -1 \times (-10 - 12) = 22$
- $C_{31} = (-1)^{3+1}|M_{31}| = (-1)^4 \begin{vmatrix} 4 & 2 \\ 3 & 1 \end{vmatrix} = 1 \times (4 - 6) = -2$
- $C_{32} = (-1)^{3+2}|M_{32}| = (-1)^5 \begin{vmatrix} 5 & 2 \\ 2 & 1 \end{vmatrix} = -1 \times (5 - 4) = -1$
- $C_{33} = (-1)^{3+3}|M_{33}| = (-1)^6 \begin{vmatrix} 5 & 4 \\ 2 & 3 \end{vmatrix} = 1 \times (15 - 8) = 7$

- $C = \begin{bmatrix} -1 & 5 & -13 \\ 0 & -11 & 22 \\ -2 & -1 & 7 \end{bmatrix}$

(iii) Find the adjoint matrix of A.

- $\text{adj}(\text{A}) = C^T$
- $\text{adj}(\text{A}) = \begin{bmatrix} -1 & 0 & -2 \\ 5 & -11 & -1 \\ -13 & 22 & 7 \end{bmatrix}$

(iv) Find the inverse of the matrix A

- Find the inverse of the matrix A
- $A^{-1} = \frac{1}{|A|} adj(A)$

$$= \frac{1}{-11} \begin{bmatrix} -1 & 0 & -2 \\ 5 & -11 & -1 \\ -13 & 22 & 7 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{11} & 0 & \frac{2}{11} \\ \frac{5}{-11} & 1 & \frac{1}{11} \\ \frac{13}{11} & -2 & \frac{-7}{11} \end{bmatrix}$$